KCF Object Tracking (PTZ)

1. Experiment Objective

The PTZ pan-tilt + mouse box selection combines the KCF (Kernelized Correlation Filter) tracking algorithm to track any object. The user drags a box with the mouse to select the target in the frame; the KCF tracker automatically learns the target features and keeps locating it, while PD control drives the pan-tilt so that the target stays at the center of the frame.

2. Experiment Principle

1. Key Terms

Term	Description
KCF (Kernelized Correlation Filter)	An online tracking algorithm based on correlation filtering; uses circulant matrices and the kernel trick to compute the target response map at high speed in the frequency domain, suitable for real-time applications
OpenCV Tracking API	The unified tracking interface provided by the OpenCV contrib module, supporting KCF/CSRT/MIL/BOOSTING/MOSSE/TLD/MEDIANFLOW and other trackers
Mouse box selection (ROI Selection)	The user drags a rectangle in the frame to specify the tracking target; the tracker extracts features from this region and keeps locating it
Visual Tracking	Detects a target and keeps locking onto it through a vision algorithm; only the pan-tilt follows, the car does not move
PTZ Pan-Tilt	A two-degree-of-freedom servo pan-tilt; s1 rotates horizontally and s2 tilts, controlled via the `/cmd_ptz_angle` topic
PD Control	Proportional-derivative control; computes the pan-tilt angle step from the target offset and the offset change rate
Deadband	When the target offset is within the deadband pixel range, the pan-tilt does not move, avoiding continuous jitter caused by tiny errors

2. Technical Architecture

```
State estimation (ekf_filter_node + camera_node) → kcf_tracker_ptz_node → /cmd_ptz_angle → firmware servo control
```

Node chain explanation:

- **ekf_filter_node**: Extended Kalman filter node that fuses IMU/odometry data for state estimation
- **camera_node**: Camera driver node that handles the image topic published by the ESP32

- **kcf_tracker_ptz_node**: KCF tracking node that processes the image at full resolution (960×720); it supports mouse box selection of the target followed by continuous tracking with the OpenCV KCF tracker, and a PD controller computes the pan-tilt angle step
- **/cmd_ptz_angle**: PTZ servo angle command topic, bridged to the ESP32 firmware by micro_ros_agent, controlling the s1 (horizontal) and s2 (tilt) servos

3. Core Topics Quick Reference

Firmware Uplink (ESP32 firmware → micro_ros_agent → ROS2 node)

Topic Name	Type	QoS	Description
`/camera/image_raw`	`sensor_msgs/Image`	BEST_EFFORT, KEEP_LAST, depth=1, VOLATILE	Raw image frames published by the ESP32 camera

Firmware Downlink (ROS2 node → micro_ros_agent → ESP32 firmware)

Topic Name	Type	QoS	Description
`/cmd_ptz_angle`	`geometry_msgs/Vector3`	RELIABLE, KEEP_LAST, depth=10	PTZ pan-tilt angle command (s1: horizontal angle, s2: tilt angle)

ROS2 Internal (inter-node communication)

Topic Name	Type	QoS	Description
`/camera/image_raw`	`sensor_msgs/Image`	BEST_EFFORT	Camera image (bridged to ROS2 by micro_ros_agent; subscribed by the tracking node)
`/cmd_ptz_angle`	`geometry_msgs/Vector3`	RELIABLE	PTZ angle command (published by the tracking node; bridged to the firmware by micro_ros_agent)

3. Experiment Steps

1. Hardware Preparation

Hardware	Requirement
PTZ version	☑
No-PTZ version	❌
Required peripherals	ESP32 camera (pan-tilt) + PTZ servo

2. Start Agent + Camera + Joint Model (Terminal 1)

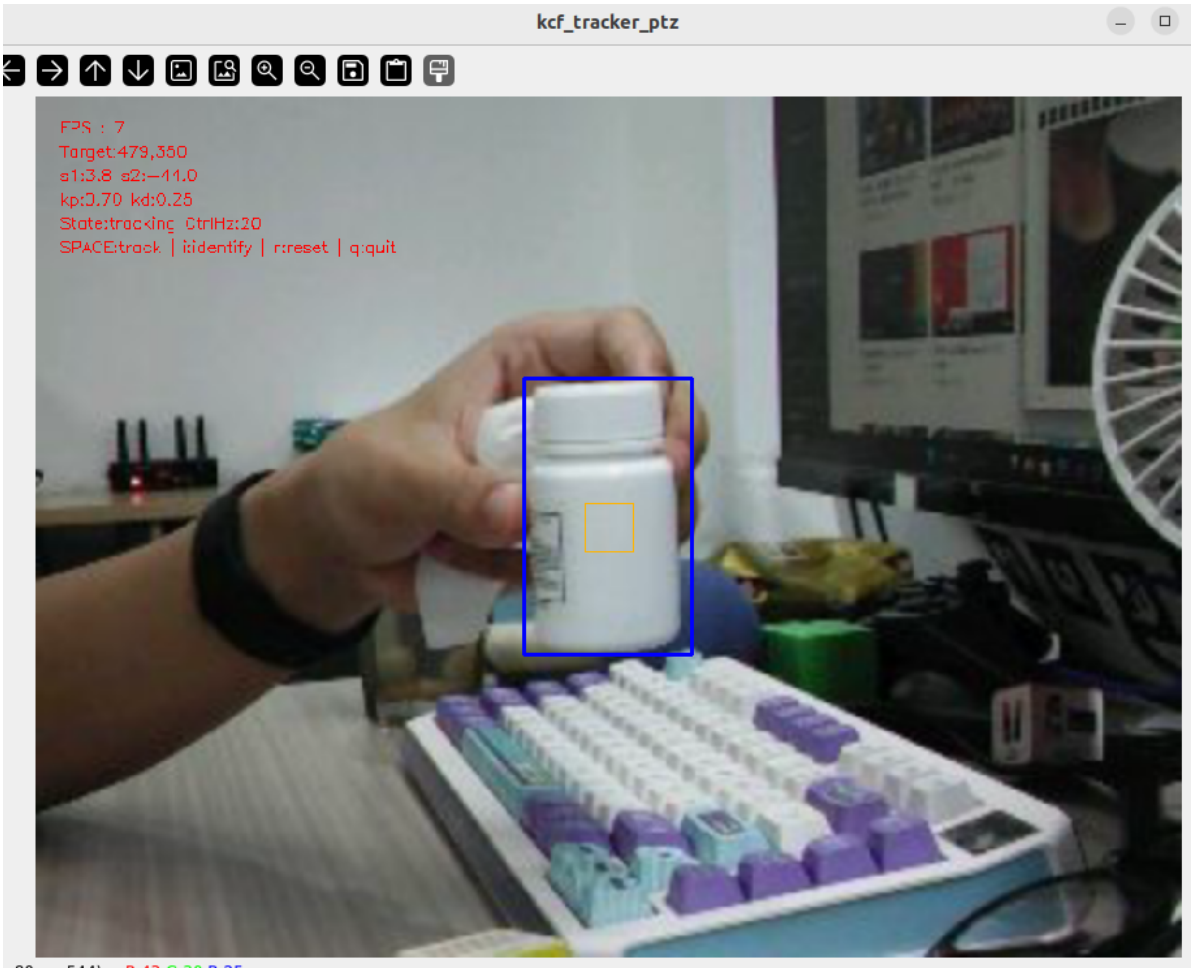
```
ros2 launch microros_v2_bringup car_base.launch.py
```

```
yahboom@yahboom:~$ ros2 launch microros_v2_bringup car_base.launch.py
[INFO] [launch]: All log files can be found below /home/yahboom/.ros/log/2026-07-28-15-04-46-269642-yahboom-579370
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [micro_ros_agent-1]: process started with pid [579411]
[INFO] [camera_driver-2]: process started with pid [579413]
[INFO] [robot_state_publisher-3]: process started with pid [579415]
[INFO] [joint_state_publisher-4]: process started with pid [579417]
[INFO] [ekf_node-5]: process started with pid [579419]
[micro_ros_agent-1] [1785222287.404236] info     | UDPv4AgentLinux.cpp | init                  | running...             | port: 8090
[robot_state_publisher-3] [INFO] [1785222288.847816851] [robot_state_publisher]: got segment base_footprint
[robot_state_publisher-3] [INFO] [1785222288.848013042] [robot_state_publisher]: got segment base_link
[robot_state_publisher-3] [INFO] [1785222288.848028942] [robot_state_publisher]: got segment bwheel_Link
[robot_state_publisher-3] [INFO] [1785222288.848037122] [robot_state_publisher]: got segment cam1_Link
[robot_state_publisher-3] [INFO] [1785222288.848043874] [robot_state_publisher]: got segment cam2_Link
[robot_state_publisher-3] [INFO] [1785222288.848050538] [robot_state_publisher]: got segment cam3_Link
[robot_state_publisher-3] [INFO] [1785222288.848057264] [robot_state_publisher]: got segment imu_frame
[robot_state_publisher-3] [INFO] [1785222288.848063747] [robot_state_publisher]: got segment laser_frame
[robot_state_publisher-3] [INFO] [1785222288.848070240] [robot_state_publisher]: got segment lfwheel_Link
[robot_state_publisher-3] [INFO] [1785222288.848076888] [robot_state_publisher]: got segment rfwheel_Link
[joint_state_publisher-4] [INFO] [1785222289.372349515] [joint_state_publisher]: Waiting for robot_description to be published on the robot_description
[camera_driver-2] [INFO] [1785222289.716161193] [esp32_ip_camera_publisher]: camera node started, camera_url: http://192.168.12.37:81/stream
[camera_driver-2] [WARN] [1785222292.705741414] [esp32_ip_camera_publisher]: Camera not opened, trying reconnect... (next retry in 1.0s)
[camera_driver-2] [INFO] [1785222294.072701944] [esp32_ip_camera_publisher]: IP camera stream opened successfully
```

3. Start the KCF Object Tracking Node (Terminal 2)

Open a second terminal and start the KCF object tracking node:

```
ros2 launch microros_v2_ptz_visual_control_pkg kcf_tracker_ptz.launch.py
```



4. Operation Instructions

After startup, the node is in the `init` state, and the main loop `spin_once` schedules image processing and keyboard polling. Image processing (`image_processing_rate_hz` = 15 Hz) and pan-tilt control (`control_rate_hz` = 20 Hz) are decoupled.

Image preprocessing

Decoding + orientation correction + scaling (`image_width`×`image_height` = 960×720) is done directly in the image callback, and the processed frame is cached for the image processing timer. The KCF tracker runs directly at the original resolution.

Target box selection (mouse interaction)

In the `init` state, drag with the left mouse button to select the target. After release, the KCF tracker is initialized automatically:

- Clip the ROI to the image boundary
- Require a width/height ≥ 10 px
- Create the `KcfTracker` (internally tries `cv.TrackerKCF_create`, falling back to the legacy version on failure)
- Initialize the tracker with the first-frame ROI; on success, switch to the `tracking` state

KCF tracking update (every frame)

The `_track()` method calls `tracker.update(frame)` every frame to obtain the new target position (ok, bbox). On success it draws a blue rectangle and updates the target center (cx, cy); on failure the frame shows "Tracking failure detected" and the target is marked lost.

PD pan-tilt control (tracking state only)

Fixed-rate 20 Hz PD control: target center pixel coordinates → normalized error → PD step → limiting (±`max_ptz_step` = 1.2°) → step smoothing (α = `ptz_step_smooth_alpha` = 0.35) → direction reversal (horizontal not reversed, tilt reversed) → angle smoothing + limiting → publish. When there is no valid target, the PD outputs nothing.

Tracker type switching

Supports dynamically modifying `tracker_type` (KCF/CSRT/MIL/BOOSTING/MOSSE/TLD/MEDIANFLOW) via `ros2 param set /kcf_tracker_ptz_node tracker_type CSRT` (or rqt); the new tracker type takes effect on the next box selection.

Shortcut keys

Key	Function	Description
Mouse drag	Select the target	Hold the left button and drag to box-select the tracking target; the KCF tracker is initialized automatically on release
Space	tracking ↔ identify	Toggles between tracking and paused control
i / I	Recognition mode	Only tracks and displays, without controlling the pan-tilt
r / R	Re-select	Clears the current tracker and returns to the init state waiting for a new box selection
q / Q / ESC	Exit	Force-exits the node process with `os._exit(0)`

On-screen overlay: The debug window shows the FPS, the target coordinates (when tracking succeeds) or "Target:None", the pan-tilt s1/s2 angles, the PD parameters (kp/kd), the current state, and the control rate. A crosshair is drawn at the frame center, the deadband is shown as an orange rectangle, and the KCF tracking box is drawn as a blue rectangle.

Safety and Edge Cases

- ROI too small is rejected: a selection smaller than 10×10 px refuses to initialize the tracker
- Tracker creation failure: automatically falls back to the legacy version when KCF is unavailable
- Tracking loss handling: when `tracker.update()` returns failure → the target is marked lost and the PD outputs no control; press `r` to re-select
- Pan-tilt angle limiting: horizontal [-90°, 90°], tilt [-90°, 20°]
- Deadband protection: the step is forced to zero when the target offset is within the deadband (20 px)
- Initial PTZ publish at startup: the initial angle (0°, -45°) is published 20 times at 0.2 s intervals

5. How to Stop

Press `Ctrl+C` in the two terminals to stop each process; it is recommended to close them in the reverse order:

```
# Terminal 2: press Ctrl+C to stop the KCF tracking node
# Terminal 1: press Ctrl+C to stop micro_ros_agent
```

4. Experiment Observations

- **init state (after startup)**: The frame streams in real time and prompts the user to drag with the mouse to box-select the tracking target; the pan-tilt stays at its initial position (0°, -45°)
- **After box selection**: The KCF tracker is initialized and automatically enters the `tracking` state; the target position is marked with a blue rectangle, and on successful tracking the frame shows the target coordinates and the kp/kd parameters
- **tracking mode**: The pan-tilt follows the target movement smoothly with PD control; KCF computes the response map in the frequency domain to update the target position in real time
- **Recognition mode (identify/paused)**: The tracker keeps running and displaying, but publishes no pan-tilt control commands

- **Tracking failure**: The frame shows the red "Tracking failure detected" prompt, the target state becomes None, and the pan-tilt stays at its last position
- **Target within the deadband**: The step is zeroed and the pan-tilt stays completely still without jitter
- **Low frame-rate scenes**: KCF is relatively tolerant of frame rate, but it is easily lost when the target moves too fast or is heavily occluded; select a larger target and move slowly
- **Switching tracker type**: Run `ros2 param set /kcf_tracker_ptz_node tracker_type CSRT` (or modify it via rqt), then press `r` to re-select for it to take effect (e.g., CSRT is more accurate but slower)

5. Program Analysis

> **Source code paths:**
>
> - Launch file: `~/microros_ws/src/microros_v2_ptz_visual_control_pkg/launch/kcf_tracker_ptz.launch.py`
> - Node file: `~/microros_ws/src/microros_v2_ptz_visual_control_pkg/microros_v2_ptz_visual_control_pkg/kcf_tracker_ptz_node.py`

1. Core Node Analysis

KcfTracker wrapper class — encapsulates multiple OpenCV trackers

```
class KcfTracker:
    """Wrapper around the built-in OpenCV trackers, supporting
KCF/CSRT/MIL/BOOSTING/MOSSE/TLD/MEDIANFLOW"""

    def __init__(self, tracker_type: str = "KCF"):
        self._type = tracker_type.upper()
        self._tracker = None
        self._working = False
        self._create()

    def _create(self) -> None:
        t = self._type
        if t == "BOOSTING":       self._tracker = cv.TrackerBoosting_create()
        elif t == "MIL":          self._tracker = cv.TrackerMIL_create()
        elif t == "KCF":          self._tracker =
self._try_create("TrackerKCF_create")
        elif t == "TLD":          self._tracker = cv.TrackerTLD_create()
        elif t == "MEDIANFLOW":   self._tracker = cv.TrackerMedianFlow_create()
        elif t == "CSRT":         self._tracker =
self._try_create("TrackerCSRT_create")
        elif t == "MOSSE":        self._tracker =
self._try_create("legacy.TrackerMOSSE_create")
        if self._tracker is None:
            self._tracker = self._try_create(f"legacy.Tracker{t}_create")
        if self._tracker is None:
            raise RuntimeError(f"Tracker {t} not available")

    def init(self, frame: np.ndarray, roi: Tuple[int, int, int, int]) -> bool:
        ok = self._tracker.init(frame, roi)
        self._working = ok if ok is not None else True
        return self._working
```

```
    def update(self, frame) -> Tuple[bool, Tuple[int, int, int, int]]:
        if not self._working:
            return False, (0, 0, 0, 0)
        ok, bbox = self._tracker.update(frame)
        if ok:
            self._working = True
            return True, tuple(int(v) for v in bbox)
        else:
            self._working = False
            return False, (0, 0, 0, 0)
```

Node initialization (`__init__`) — state machine + image/control dual timers

```
class KcfTrackerPtzNode(Node):

    def __init__(self) -> None:
        super().__init__(DEFAULT_NODE_NAME)
        self._img_lock = threading.Lock()
        self._tgt_lock = threading.Lock()

        self._latest_frame: Optional[np.ndarray] = None
        self._state = STATE_INIT         # "init" / "identify" / "tracking"
        self._select_flags = False        # mouse box-selection state
        self._roi_init = (0, 0, 0, 0)     # ROI selected by the mouse

        self._tracker: Optional[KcfTracker] = None
        self._tracker_ready = False
        self._has_target = False
        self._target_x: Optional[float] = None
        self._target_y: Optional[float] = None

        self._declare_params()
        self._load_params()
        self._h_deg = self._init_h_deg; self._p_deg = self._init_p_deg

        # Create the image subscription (BEST_EFFORT QoS)
        self._sub = self.create_subscription(
            Image, self._img_topic, self._on_image,
            QoSProfile(history=QoSHistoryPolicy.KEEP_LAST, depth=1,
                       reliability=QoSReliabilityPolicy.BEST_EFFORT,
                       durability=QoSDurabilityPolicy.VOLATILE),
        )
        self._pub = self.create_publisher(Vector3, self._ptz_topic, 10)

        # Image processing timer (15 Hz)
        self._img_timer = self.create_timer(
            1.0 / max(self._img_rate, MINIMUM_TIMER_RATE_HZ),
self._process_image,
        )
        self._ctrl_timer = None; self._rebuild_ctrl_timer(False)

        # Mouse callback + keyboard timer
        if self._show_win:
            self._init_win()
            self._key_timer = self.create_timer(0.05, self._poll_keys)
```

```
        # Initial PTZ angle publishing
        if self._pub_init: self._start_init_ptz()

        self.get_logger().info(
            f"KCF tracker started. Drag ROI → SPACE to track. "
            f"ctrl={self._ctrl_rate:.1f}Hz")
```

Mouse box selection initializes the tracker (`_on_mouse` + `_init_tracker`)

```
def _on_mouse(self, event, x, y, flags, param):
    if event == cv.EVENT_LBUTTONDOWN:
        self._state = STATE_INIT; self._select_flags = True
        self._mouse_xy = (x, y); self._ms_start = (x, y); self._ms_end = (x, y)
    elif event == cv.EVENT_MOUSEMOVE and self._select_flags:
        self._ms_start = self._mouse_xy; self._ms_end = (x, y)
        self._roi_init = (self._ms_start[0], self._ms_start[1],
                          self._ms_end[0], self._ms_end[1])
    elif event == cv.EVENT_LBUTTONUP:
        self._select_flags = False
        self._init_tracker(frame_hint=None)

def _init_tracker(self, frame_hint=None) -> None:
    x1, y1, x2, y2 = self._roi_init
    if x1 == x2 or y1 == y2: return
    rx = min(x1, x2); ry = min(y1, y2)
    rw = abs(x2 - x1); rh = abs(y2 - y1)
    if rw < 10 or rh < 10: return           # reject a ROI that is too small
    rx = int(clamp(rx, 0, self._img_w - 1))
    ry = int(clamp(ry, 0, self._img_h - 1))
    rw = int(min(rw, self._img_w - rx))
    rh = int(min(rh, self._img_h - ry))
    roi = (rx, ry, rw, rh)

    with self._img_lock:
        frame = self._latest_frame
        if frame is None: return
        init_frame = frame.copy()

    # Create and initialize the KCF tracker
    self._tracker = KcfTracker(self._trk_type)
    self._tracker.init(init_frame, roi)
    self._tracker_ready = True
    self._reset_pd()
    self._state = STATE_TRACKING
    self.get_logger().info(f"Tracker ready. ROI={roi}")
```

KCF tracking update (`_track`) — calls tracker.update() every frame

```
def _track(self, frame: np.ndarray) -> Tuple[bool, float, float]:
    if self._select_flags and hasattr(self, '_ms_start') and hasattr(self,
'_ms_end'):
        cv.rectangle(frame, self._ms_start, self._ms_end, (0, 255, 0), 2)

    if self._state == STATE_INIT:
        return False, 0.0, 0.0
    if not self._tracker_ready or self._tracker is None:
```

```
        return False, 0.0, 0.0

    ok, bbox = self._tracker.update(frame)
    if not ok:
        cv.putText(frame, "Tracking failure detected", (100, 80),
                   cv.FONT_HERSHEY_SIMPLEX, 0.75, (0, 0, 255), 2)
        with self._tgt_lock: self._has_target = False
        return False, 0.0, 0.0

    x, y, w, h = bbox
    cx, cy = x + w / 2.0, y + h / 2.0
    cv.rectangle(frame, (x, y), (x + w, y + h), (255, 0, 0), 2, 1)

    with self._tgt_lock:
        self._has_target = True
        self._target_x = float(cx); self._target_y = float(cy)
    return True, float(cx), float(cy)
```

PD pan-tilt control (`_ctrl_tick`) — shares the same PD algorithm as the pose tracking node

```
def _ctrl_tick(self) -> None:
    if self._state != STATE_TRACKING: return
    with self._tgt_lock:
        tx, ty, ok = self._target_x, self._target_y, self._has_target
    if not ok or tx is None or ty is None: return

    icx, icy = self._img_w * 0.5, self._img_h * 0.5
    pex, pey = tx - icx, ty - icy
    cx, cy = abs(pex) < self._db_x, abs(pey) < self._db_y
    if cx: pex = 0.0
    if cy: pey = 0.0

    n = time.perf_counter()
    dt = n - self._prev_ctrl_t if self._prev_ctrl_t > 0.0 else 1.0 / max(...)
    dt = max(dt, 1e-3)
    nex, ney = pex / icx, pey / icy
    dx, dy = (nex - self._prev_err_x) / dt, (ney - self._prev_err_y) / dt
    rx = clamp((self._kp * nex + self._kd * dx) * self._max_s,
               -self._max_s, self._max_s)
    ry = clamp((self._kp * ney + self._kd * dy) * self._max_s,
               -self._max_s, self._max_s)
    sx = (1.0 - self._step_a) * self._last_sx + self._step_a * rx
    sy = (1.0 - self._step_a) * self._last_sy + self._step_a * ry
    self._h_deg = clamp(...)
    self._p_deg = clamp(...)
    self._pub_ptz()
```

2. Key Parameters

> **Parameter definition file:**
> `~/microros_ws/src/microros_v2_ptz_visual_control_pkg/launch/kcf_tracker_ptz.launch.py`

Parameter	Type	Default Value	Description
`pd_kp` / `pd_kd`	float	`0.7` / `0.25`	PD proportional/derivative gains
`control_rate_hz`	float	`20.0` Hz	Pan-tilt control rate
`image_processing_rate_hz`	float	`15.0` Hz	Image processing rate
`deadband_x` / `deadband_y`	float	`20.0` / `20.0` px	Horizontal/vertical deadband
`max_ptz_step`	float	`1.2`°	Maximum single angle step
`ptz_step_smooth_alpha`	float	`0.35`	Step EMA smoothing coefficient
`ptz_smooth_alpha`	float	`0.85`	Pan-tilt angle EMA smoothing coefficient
`tracker_type`	str	`KCF`	Tracker type: KCF/CSRT/MIL/BOOSTING/MOSSE/TLD/MEDIANFLOW
`image_width` / `image_height`	int	`960` / `720` px	Processing image resolution (full resolution)
`window_width` / `window_height`	int	`960` / `720` px	Debug window size
`initial_horizontal_deg` / `initial_pitch_deg`	float	`0.0`° / `-45.0`°	Initial pan-tilt angles

6. Frequently Asked Questions

Problem	Cause	Solution
Unstable tracking	Low frame rate + KCF is sensitive to occlusion/scale changes	Box-select a larger target, move slowly, avoid occlusion; or switch to CSRT (more accurate)
Tracker initialization failure	OpenCV contrib not installed or an incompatible version	Make sure the `python3-opencv` contrib module is installed, or switch `tracker_type` to an available type
No automatic recovery after target loss	KCF has no loss-recovery mechanism	Press `r` to box-select the target again
Tracking box drifts to the background	The background texture resembles the target	Make sure the target texture inside the ROI is distinctive when selecting, and avoid large solid-color or textureless objects